Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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Valuation of a stock

(Chapter 13)

Price of a Stock

We all know **Apple** to be a successful company

- ✓ any idea how much a stock of Apple would cost?
  - check it out

- ✓ and check out the growth rate: on 1st Aug 1997 it was selling for 65.8 cents

- ✓ it really only took off when first i-Phone was announced in 2007

- ✓ they are now the company with the largest market capitalization in the world: $3.47 trillion!
  - it was $2.4 trillion 2 ys ago

Price of a Stock

- ✓ Ever heard of **Berkshire Hathaway Inc.**?
  - it’s Warren Buffett’s investment company
  - Warren Buffett is one of the richest man on earth, he’s been running the company for 50+ years
- ✓ Buying a share in Berkshire Hathaway seems like a good investment.
  - want to buy? Check out the price
- ✓ It’s one of the few financial companies in the US among the top 20 by market cap

Computing the Price of a Stock

- ✓ The principle of valuing common stock is no different from valuing debt securities:

  1. Determine the cash flows

  2. Discount them to the present

  3. Price them at their PV

- ✓ But … the pricing of a stock is much **harder** in practice than it is for bonds. Why?

Computing the Price of a Stock

Q: what are the two items that compose a stock cash flow? (*i.e.* how do you make money by owning a stock?)

Are they both certain?

Computing the Price of a Stock

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Are they both certain?

A:

1. **Dividends** (share of company's profits distributed to stockholders)

2. **Future sales price**

No, neither one is certain …

… and different stockholders may have different information about the stock and thus they have different valuations

The One-Period Valuation Model

✓ Take the simplest model, just take the expected dividend and the expected price over the next year.

$$P_0 = \frac{Div_1}{1+k_e} + \frac{P_1}{1+k_e}$$

where:

- P_0 is the price today
- P_1 is the (forecasted) price next year
- Div_1 is the (expected) dividend next year
- k_e: rate for discounting stocks
  - works like YTM, but *usually it's higher …*
  - because stocks are riskier, so you discount future cash flows more!

The One-Period Valuation Model

- ✓ Suppose you see a stock trading at $50 paying $0.16 in dividend per year
- ✓ You read an analyst forecast that price is expected to rise to $60 next year
- ✓ You decide you want to have 12% to be compensated for risk

Q: Find the price using your compensation for risk P_0^*.

Should you buy?

The One-Period Valuation Model

$$P_0^* = \frac{\$0.16}{(1+0.12)} + \frac{\$60}{(1+0.12)} = \$53.71$$

- ✓ if market price is lower than (or equal to) your value ($P_0 \leq P_0^*$ or also: $\$50 \leq \$53.71$): **buy**
- ✓ if market price is higher than your value ($P_0 > P_0^*$ or also: $\$50 > \$53.71$): do **not** buy
- ✓ Obviously, other investors may have different $k_e$or P_0^* … or the model is wrong

The Generalized Dividend Valuation Model

- ✓ Using the same logic, extend to n periods:

$$P_0 = \frac{Div_1}{1+k_e} + \frac{Div_2}{(1+k_e)^2} + \cdots + \frac{Div_n}{(1+k_e)^n} + \frac{P_n}{(1+k_e)^n}$$

- ✓ but as *n* grows, last item in the summation gets smaller (higher discounting). So, end up with:

$$P_0 = \sum_{t=1}^{\infty} \frac{Div_t}{(1+k_e)^t}$$

- ✓ <u>Note:</u>
  - ∞ is the end of summation … firms are supposed to be infinitely lived
  - no final $\frac{P_n}{(1+k_e)^n}$ in the second expression, bcs for n large enough it is a very low number, so that you don't need to forecast P_n
  - just PV of *all* future dividends matter (*<u>fundamentals</u>*)

- ✓ But it is "difficult" to forecast an infinite stream of payments …

The Gordon Growth Model

- ✓ Same as the previous model, but just *assume* that dividend grow at a constant rate g:

$$Div_{t+1} = (1+g)Div_t$$

- ✓ Then price of stock is:

$$P_0 = \frac{Div_0(1+g)}{(1+k_e)} + \frac{Div_0(1+g)^2}{(1+k_e)^2} + \cdots + \frac{Div_0(1+g)^n}{(1+k_e)^n}$$

- ✓ With some *algebra* we can show that:

$$P_0 = \sum_{t=1}^{\infty} \frac{Div_t}{(1+k_e)^t} = \boxed{\frac{Div_0(1+g)}{k_e - g}} \quad or \quad \boxed{P_0 = \frac{Div_1}{k_e - g}}$$

- ✓ How convenient!
  - we don't need to forecast an infinite stream of payments
  - just get **an estimate of g** and of k_e ("*required rate of return*")

The Gordon Growth Model: Example

Let's get back to previous example, with $P_0 = \$53.71$

- ✓ What is the implied dividend growth (g) with a required rate of return (k_e) of 10%?

$$P_0 = \frac{Div_0(1+g)}{k_e - g} = \frac{Div_1}{k_e - g} \Rightarrow \$53.71 = \frac{Div_1}{(0.1 - g)}$$

- ✓ But conveniently by definition $Div_1 = (1+g)Div_0$, so:

$$\$53.71 = \frac{Div_0(1+g)}{(0.1-g)} = \frac{\$0.16\,(1+g)}{(0.1-g)}$$

- ✓ *Solve for g:*

$$\$53.71 \cdot (0.1 - g) = \$0.16(1+g)$$

$$\Rightarrow 0.1 \cdot \$5.371 - \$53.71g = \$0.16 + 0.6g$$

$$\Rightarrow \$5.371 - \$0.16 = g(\$53.71 + \$0.16)$$

$$\boldsymbol{g = 9.67\%}$$

The Gordon Growth Model: Assumptions

1) Do dividends grow at a constant rate forever?

- No, but if they grow constantly for some time the model is reasonable …
- … after "some" time? Again, cash flows 20 years from now are *very difficult to predict*, but they are also very small when discounted

2) Note another assumption: $\boldsymbol{k_e > g}$

- otherwise, we can't simplify to $P_0 = \frac{D_1}{k_e - g}$ **…**
- but this is also reasonable: if $k_e < g$ Gordon formula no longer valid: why?
- at home: what happens if $k_e = g$?

The Gordon Growth Model: Interpretation

- ✓ How about companies (the majority) **that do not pay dividends** to shareholders?
  - ○ Suppose Google Dividend=0, is its price=0?
  - ○ No! Investors buy Google because expect the company's profits ↑
- ✓ Even if earnings are not distributed, they are re-invested within the firm …
  - ○ … at some point in the future, shareholders will get paid …
  - ○ … at an increased value of the firm
    - ▪ (suppose for sake of simplicity, that re-investment of earnings is at the same rate of return k_e)
- ✓ So, the "unpaid" dividends still "belong" to the stockholder
- ✓ Dividends can be loosely interpreted as <u>any cash flow stockholders are entitled to</u>

EXERCISE: Volkswagen Dieselgate

- ✓ It's Jan 2015. Suppose Volkswagen announced dividends for €4.8 per share this year, growing at 10% forever.

- ✓ However in September 2015, following the ''dieselgate'' scandal, VW announces:
  - it will only pay dividends of €0.11 in 2016
  - from 2017 it will pay €4 and growing at 8% thereafter

Q: what is VW expected price today (2015) before and after the dieselgate happens? What is %ΔP?

Assume $k_e = 12\%$ (round to nearest integer)

EXERCISE: Volkswagen Dieselgate

- ✓ a (before the dieselgate):

$$P_{2015,a} = \frac{Div_{2016,a}}{k_e - g_a} = \frac{€4.8 \times 1.1}{12\% - 10\%} = €264$$

- ✓ b (after the dieselgate):

$$P_{2015,b} = \frac{Div_{2016,\,b}}{1+k_e} + \frac{P_{2016,\,b}}{1+k_e} \Longrightarrow P_{2015,b} = \frac{Div_{2016,b}}{1+k_e} + \left[\overbrace{\frac{Div_{2017,b}}{k_e - g_b}}^{=P_{2016,b}}\right] \times \frac{1}{1+k_e}$$

$$P_{2015,b} = \frac{€0.11}{1.12} + \left[\frac{€4}{0.12 - 0.08}\right] \times \frac{1}{1.12} \Longrightarrow P_{2015,b} = €89.4$$

$$\%\Delta P_{2015} = \frac{\left(P_{2015,b} - P_{2015,a}\right)}{P_{2015,b}} = \frac{(€89.4 - €264)}{€89} = -195.4\%$$

- ○ Note that numerator of $P_{2016,b}$ is $Div_{2017,b}$, not $Div_{2016,b}(1+g_b)$, because in this example $Div_{2016,b}(1+g_b) \neq Div_{2017,b}$, cannot apply $Div_{2016,b}(1+g_b)$

Price Earnings Valuation

- ✓ The **price earnings ratio** (PE ot P/E)
  - PE is a measure of <u>how much the market is willing to pay for $1.00 of company's earnings</u>: $PE = \frac{P}{E}$
  - widely watched measure
  - easy to calculate and avoids dividends (often they are not distributed) by looking at earnings

- ✓ Trivially then, by definition:

This is the PE

$$Price = \frac{P}{E} \times Earnings$$

  - where E are the earnings per share (EPS) and, obviously, P is the share price

- ✓ If PE is 20 it means that investors are willing to pay $20 today for each $ of company's earnings per share

Price Earnings Valuation

$$Price = PE \times Earnings$$

- ✓ *E* is **expected earnings per share**
  - since future earning are unknown, in practice it is common to use *actual* earnings per share from the last 12-months

- ✓ Is the PE high or low? It depends on relatively to what!

- ✓ As a reference the PE can also be taken from **industry average**. Why?
  - because it is expected that a firm PE will *equal* the industry average PE in the long run

Price Earnings Valuation

- ✓ A PE of a firm higher than industry-average $\left(PE_f > PE_j\right)$ has two interpretations ($f = firm$; $j = industry$):
  - in a "high PE" company, *earnings* are **expected to rise** in the future compared to low PE companies
  - the company is **low risk**, hence the market pays a premium for its earnings
- ✓ Growth/Glamour stocks are those with high PE
- ✓ A "low PE" $\left(say\text{: } PE_f < PE_j\right)$ means:
  - firm will have **lower** future *earnings* that are **already** reflected in the stock price (note P/E is obtained by dividing by the *higher current* earnings)
  - that the firm is **quite risky** and thus the stockholders require more compensation (in form of current earnings) to hold this stock

Price Earnings Valuation

Example

- ✓ Firm A’s stock currently earns \$1.55 per share and costs at the moment \$24.01. Its P/E is in line with industry

- ✓ Firm B’s stock currently earns \$1.96 per share. Assume that it belongs to the same industry as firm A

Q.: How much we would expect it is valued today? Assume that it has a PE in line with that of the industry

Price Earnings Valuation

Example

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Q.: How much we would expect it is valued today? Assume that it has a PE in line with that of the industry

A: P/E (firm A)=\$24.01/\$1.55 = 15.49

P/E (firm B)= \$1.96 $\times$ 15.49 = \$30.36

Errors in Valuation

- ✓ Wow! We can price stocks … let's go make some money!
- ✓ note that all the models require to compute **expected earnings/dividends/growth rates**
- ✓ sometimes far ahead in the future
- ✓ and *small errors* in evaluating each can lead to *big mistakes* (read: big losses!)
- ✓ For example, in the Gordon model we need two items: $\boldsymbol{g}$ and k_e, errors could depend on both $\boldsymbol{g}$ and k_e

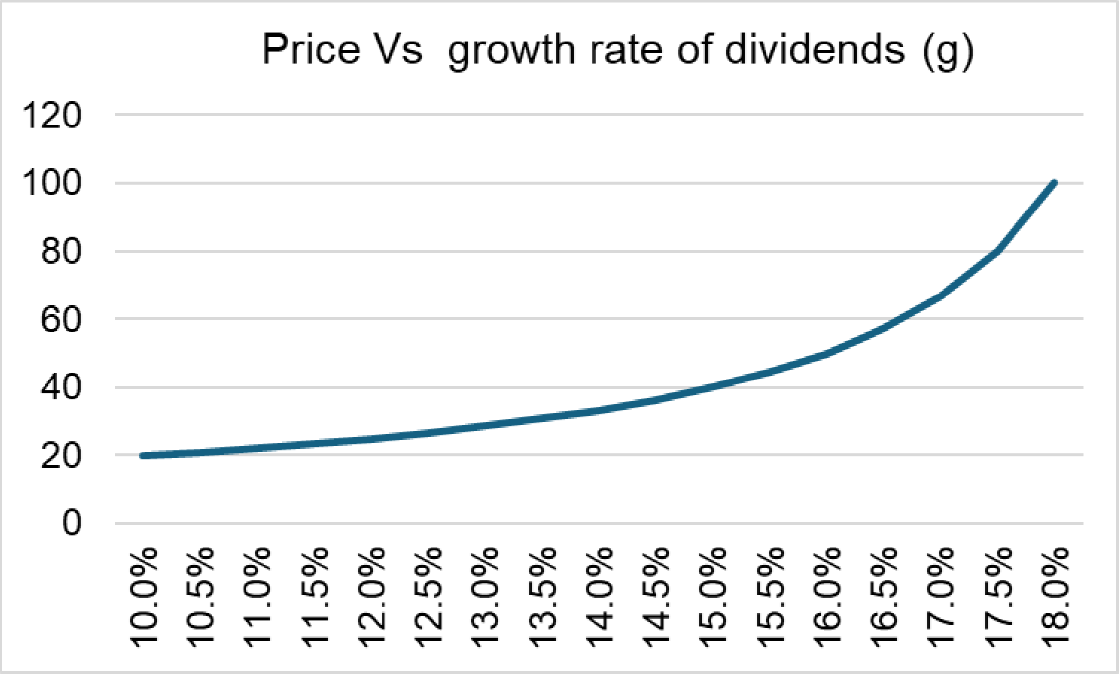
The Gordon Growth Model: Example

Let's see how assumptions on g and on k_e may create large evaluation errors

Example:

- dividend is \$2
- fix k_e=0.2 and let g **vary**

$$P_0 = \frac{Div_1}{k_e - g}$$



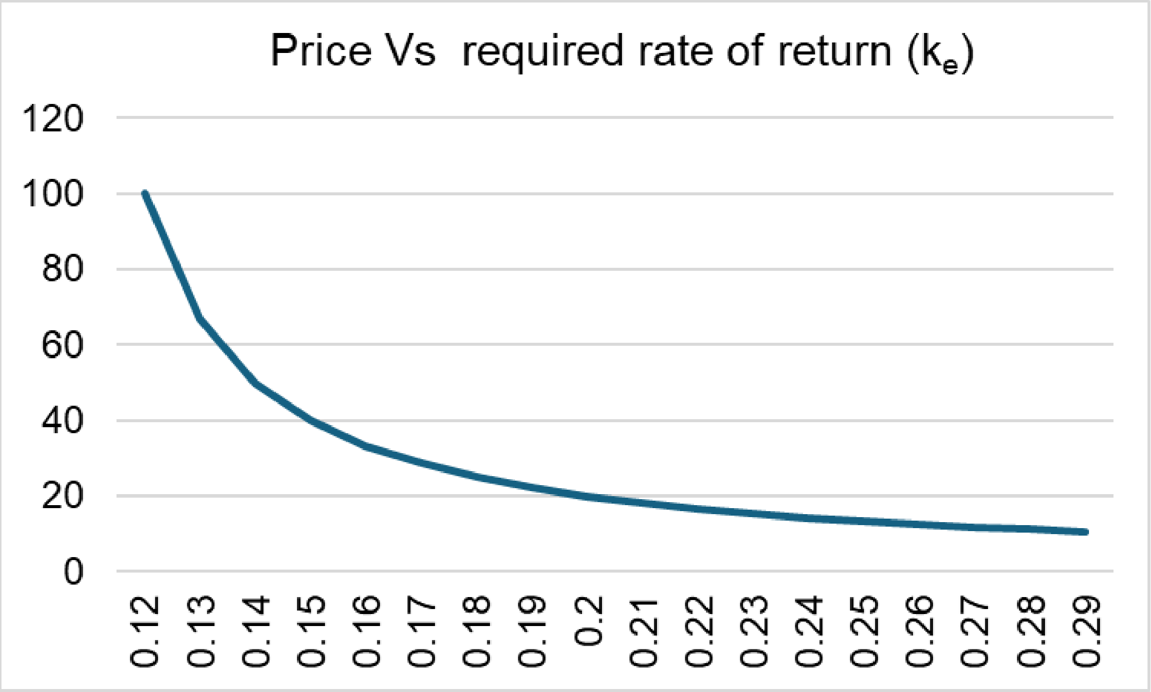
- different (projected) growth rates result in very different valuations of the stock!

The Gordon Growth Model: Example

Same example:

- dividend is \$2
- fix g=0.11 and let k_e **vary**

$$P_0 = \frac{Div_1}{k_e - g}$$



- different (projected) required rates of return imply very different valuations of the stock!

Errors in Valuation: *g*

Table 13.1 Stock Prices for a Security with D_0 = \$2.00, k_e = 15%, and Constant Growth Rates as Listed

Growth (%)	Price (\$)
1	14.43
3	17.17
5	21.00
10	44.00
11	55.50
12	74.67
13	113.00
14	228.00

ΔP (and also mistakes) getting bigger as *g* grows

The 2007–2009 Financial Crisis and the Stock Market

- ✓ The Financial Crisis, which started in August 2007, was the start of one of the worst *bear*[(1)] markets in history

Q: What may contribute to low prices in a recession according to Gordon model?

(1): bear markets if stock prices fall, bull markets if they rise

The 2007–2009 Financial Crisis and the Stock Market

✓ The Financial Crisis, which started in August 2007, was the start of one of the worst *bear*[1] markets in history

Q: What may contribute to low prices in a recession according to Gordon model?

1. g is revised **downward** as expectation of future growth for US companies are revised <u>downward</u>
2. k_e is also **higher** due to <u>higher uncertainty</u>
   - discount the future more

✓ both contributed to **a fall** in *price of stocks*

(1): bear markets if stock prices fall, bull markets if they rise

How the Market sets Prices

✓ Suppose there are three investors interested in a stock

1. <u>You</u>: *do not know much* about the company, just read on WSJ that dividends are expected to be \$2 and grow at 3%. Uncertainty requires $\boldsymbol{k_e}$**=15%**.

2. <u>Jennifer</u>: she knows the industry. Feels *confident* the estimates are quite accurate. Lower $\boldsymbol{k_e}$**=12%**.

3. <u>Bud</u>: is *dating* the CEO of the company. He knows what the company's plans are. He only requires $\boldsymbol{k_e}$**=7%**.

How the Market sets Prices

- ✓ Applying the Gordon model yields the following prices:

Investor	Discount Rate	Stock Price
You	15%	$16.67
Jennifer	12%	$22.22
Bud	7%	$50.00

- ✓ The prices reflect the willingness-to-pay (maximum acceptable price) of each investor
- ✓ In a market auction, Bud would get the stock. He would bid just above Jennifer's price

How the Market sets Prices

- ✓ Two important lessons about competitive auction markets:

  1. the price is set by the buyer with the highest *w*illingness *to p*ay (but *not necessarily* at P=*w*.*t*.*p*.)

  2. the price is set by the buyer who can take best advantage of the asset

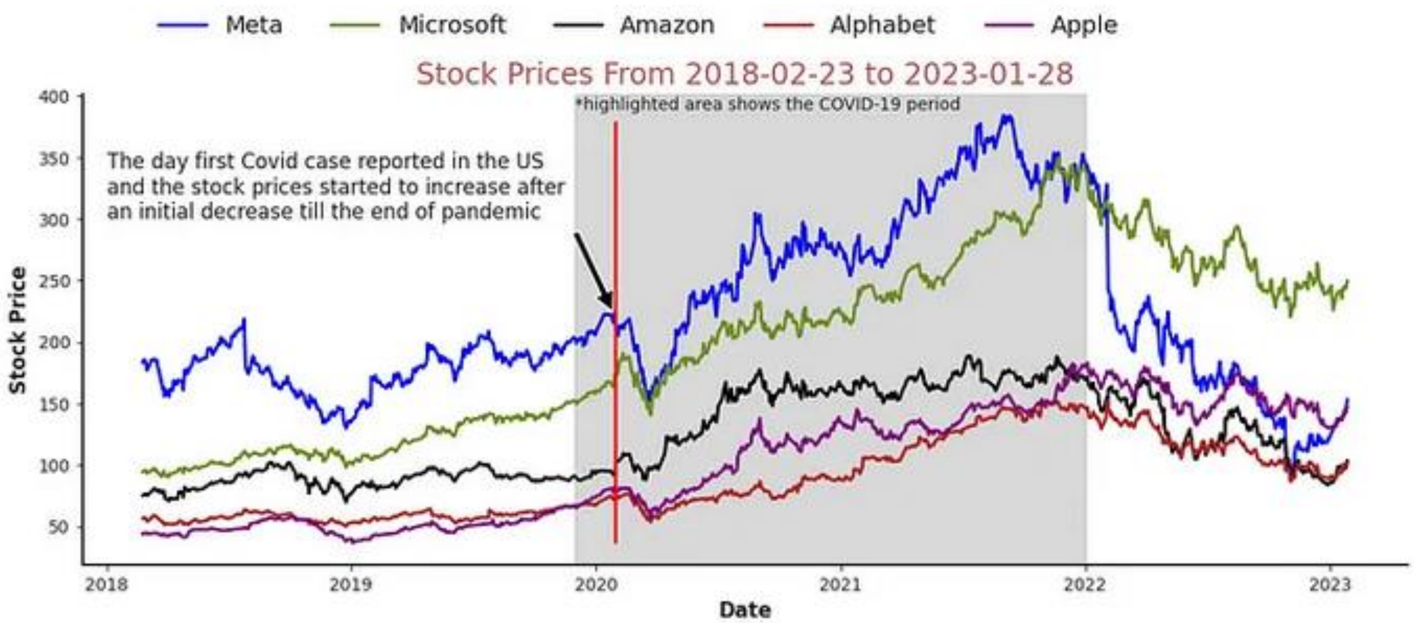
- ✓ Information matters!! This is why stock prices fluctuate widely on news reports

- ✓ Price may also be due to <u>overoptimism</u> driven by <u>market euphoria</u> (after all, Bud is in love…):
  - if the expectations are wrong, we have a **bubble** (prices too high) that will burst and create damage

How the Market sets Prices

- ✓ Given different evaluations (4 ex.: *g* and *k* in the Gordon model), stock analysts are *seldom* certain that their stock price projections are accurate

- ✓ So, should we be skeptical on investing in stock markets?
  - no, because short term fluctuations in **stock prices** are **normal**
  - in the long-term stock **prices will reflect true earnings of the firm**



Next class

We will discuss derivatives:

Forwards and Futures